

Problem Set 9 (35 pts)

Due at 11:59pm

Friday, November 7

Exercises are from <https://aimacode.github.io/aima-exercises/> after fixing numerous typos in their descriptions.

1.(13 pts) (Exercise 13.8) You are given the full joint distribution below.

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	0.108	0.012	0.072	0.008
$\neg$ <i>cavity</i>	0.016	0.064	0.144	0.576

Calculate the following probability distributions:

(a) (3 pts) $P(\textit{Toothache})$.

(b) (3 pts) $P(\textit{Cavity})$.

(c) (3 pts) $P(\textit{Toothache} \mid \textit{cavity})$.

(d) (4 pts) $P(\textit{Cavity} \mid \textit{toothache} \vee \textit{catch})$.

2.(10 pts) (Exercise 13.16) Consider two medical tests, A and B, for a virus. Test A is 95% effective at recognizing the virus when it is present, but has a 10% false positive rate (indicating that the virus is present, when it is not). Test B is 90% effective at recognizing the virus, but has a 5% false positive rate. The two tests use independent methods of identifying the virus. The virus is carried by 1% of all people. Say that a person is tested for the virus using only one of the tests, and that test comes back positive for carrying the virus. Which test returning positive is more indicative of someone really carrying the virus? Justify your answer mathematically.

[Let V be the statement that the patient has the virus, and let v and $\neg v$ stand for $V = \text{True}$ and $V = \text{False}$, respectively. Let A and B be respectively the statements that the outcomes of the medical tests A and B are positive, and $a, \neg a, b, \neg b$ be their truth values.]

3.(12 pts) (Exercise 13.18) After your yearly checkup, the doctor has bad news and good news. The bad news is that you tested positive for a serious disease and that the test is 99% accurate (i.e., the

probability of testing positive when you do have the disease is 0.99, as is the probability of testing negative when you don't have the disease). The good news is that this is a rare disease, striking only 1 in 10,000 people of your age. Why is it good news that the disease is rare? What are the chances that you actually have the disease?

[Use the random variable *Positive* to represent the test outcome, with values *positive* and $\neg$ *positive*. Use the random variable *Disease* with values *disease* for having the disease and $\neg$ *disease* for not.]